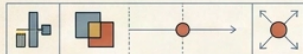


BUILDING BLOCKS OF MODERN SYSTEM DESIGN

A bottom-up, first-principles guide to scalable distributed systems.

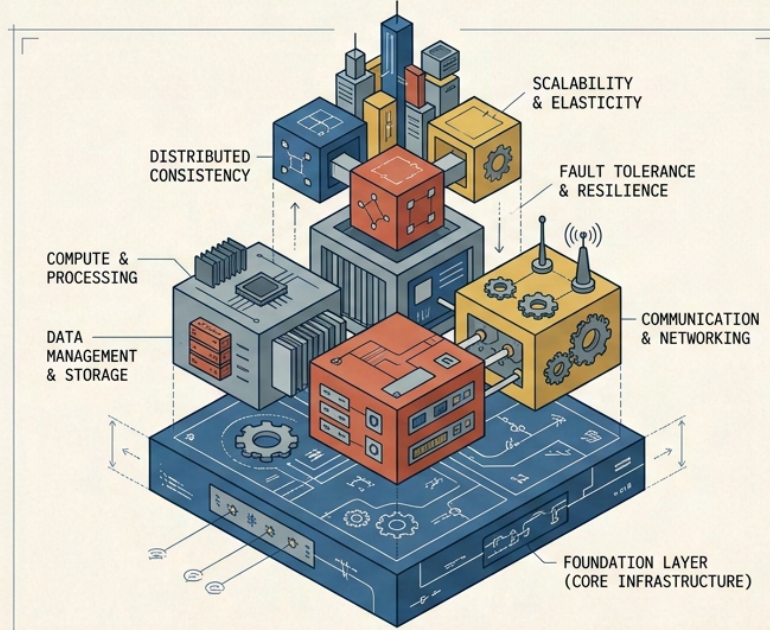
PRINCIPLES OF DESIGN

- FIRST-PRINCIPLES APPROACH
- MODULARITY & ENCAPSULATION
- DECOUPLING & LOOSE COUPLING
- RESILIENCE BY DESIGN
- SCALABILITY (HORIZONTAL/VERTICAL)



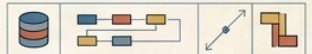
PRINCIPLES OF DESIGN

- FIRST-PRINCIPLES APPROACH
- MODULARITY & ENCAPSULATION
- DECOUPLING & LOOSE COUPLING
- RESILIENCE BY DESIGN
- SCALABILITY (HORIZONTAL/VERTICAL)



KEY COMPONENTS

- DISTRIBUTED DATABASES
- MESSAGE QUEUES & BROKERS
- LOAD BALANCERS
- SERVICE MESH
- CONSENSUS ALGORITHMS



KEY COMPONENTS

- DISTRIBUTED DATABASES
- MESSAGE QUEUES & BROKERS
- LOAD BALANCERS
- SERVICE MESH
- CONSENSUS ALGORITHMS

MASTER THE BRICKS TO BUILD THE SYSTEM

THE PATTERN

Most complex system design problems share recurring patterns, regardless of specific requirements.

THE SOLUTION

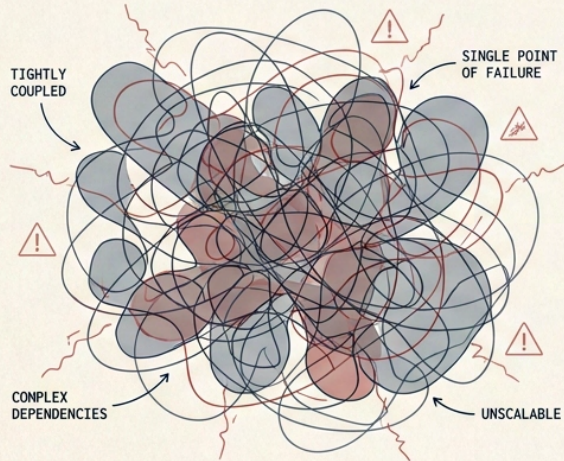
We group these recurring patterns into isolated, core building blocks.

THE ADVANTAGE

By mastering the design of each block independently, we can efficiently reuse them to construct reliable distributed systems without reinventing the wheel.

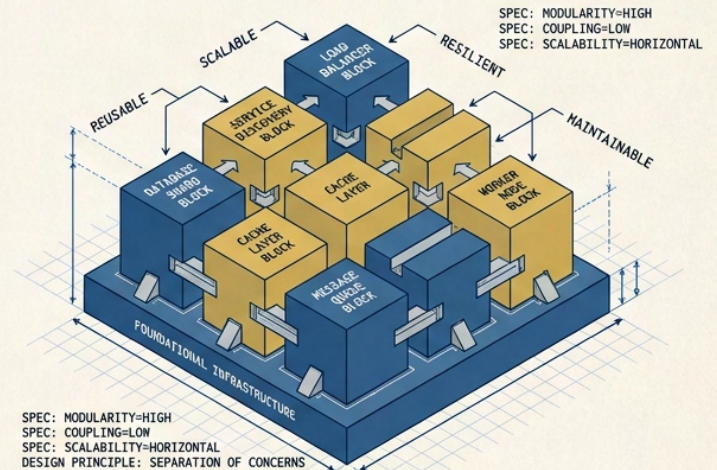
THE MONOLITH

TECHNICAL DIAGRAM: LEGACY SYSTEM



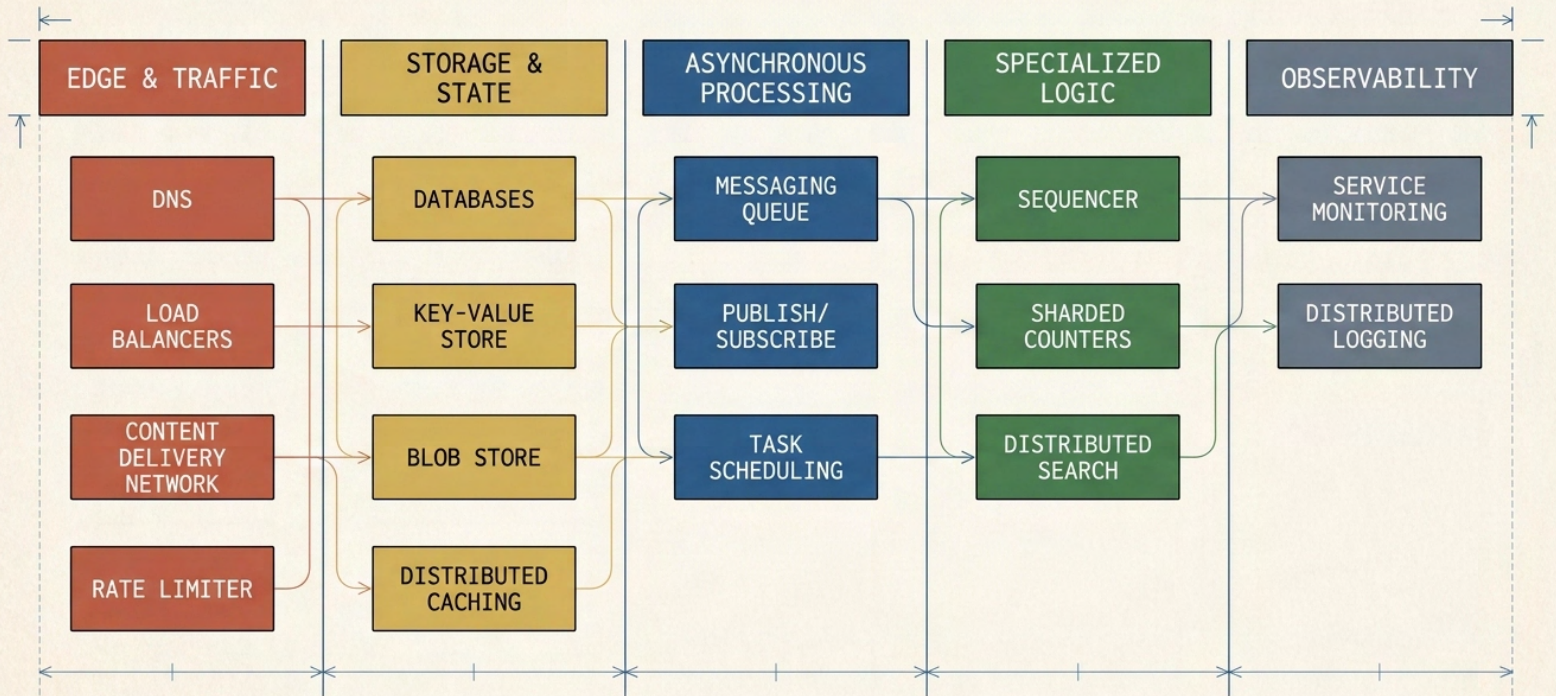
THE BOTTOM-UP APPROACH

TECHNICAL DIAGRAM: MODULAR SYSTEM ARCHITECTURE



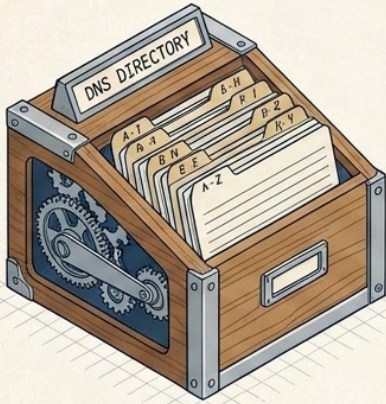
THE ARCHITECTURE ATLAS

Topological Arrangement: Components are mapped by dependency. Mastery moves left to right.



DOMAIN NAME SYSTEM (DNS)

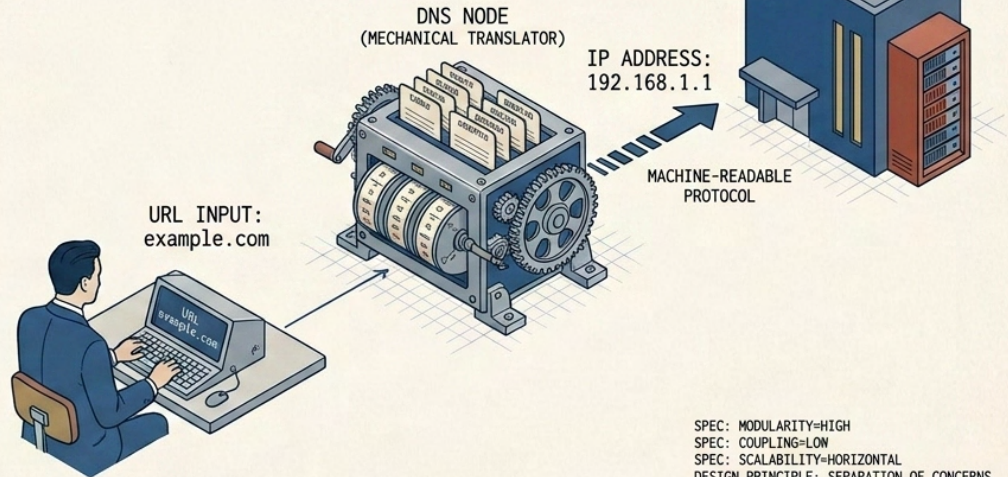
EDGE & TRAFFIC LAYER



CORE FUNCTION:

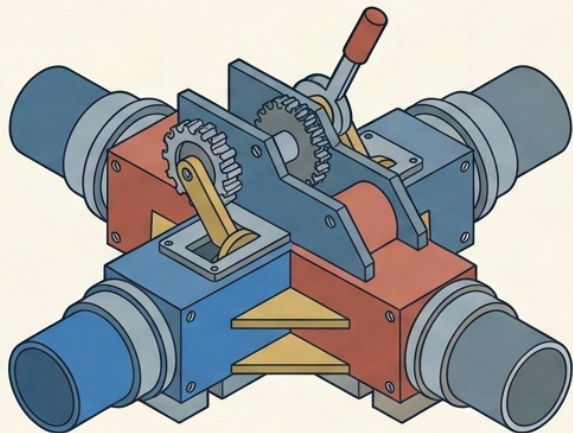
A hierarchical and distributed naming system for computers connected to the internet. Translates human-readable domain names into machine-readable internet protocols.

FIRST PRINCIPLE:
The internet's address translator.



LOAD BALANCERS

EDGE & TRAFFIC LAYER

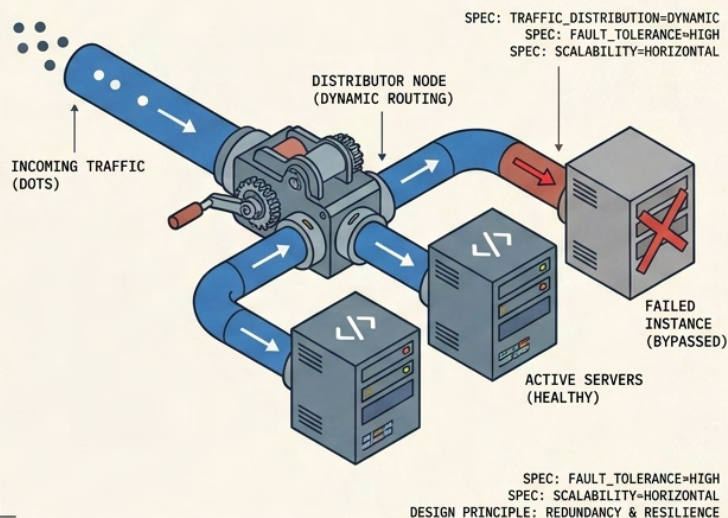


CORE FUNCTION:

Distributes incoming client requests across a pool of servers. Prevents individual servers from becoming overloaded and intelligently routes traffic away from failed instances.

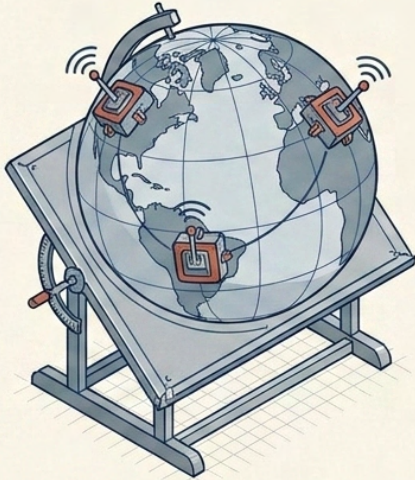
FIRST PRINCIPLE:

The dynamic traffic distributor.



CONTENT DELIVERY NETWORK (CDN)

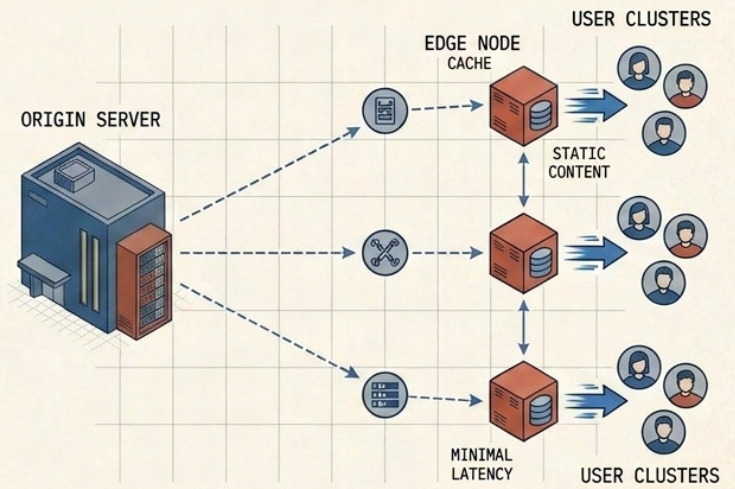
EDGE & TRAFFIC LAYER



CORE FUNCTION:

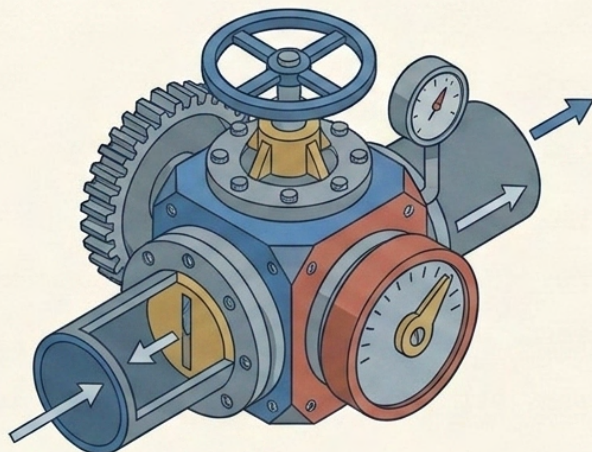
Stores and delivers viral content (videos, images, webpages). Efficiently serves end users from the closest node, drastically reducing latency and alleviating the burden on central data centers.

FIRST PRINCIPLE:
Proximity-based static delivery.



RATE LIMITER

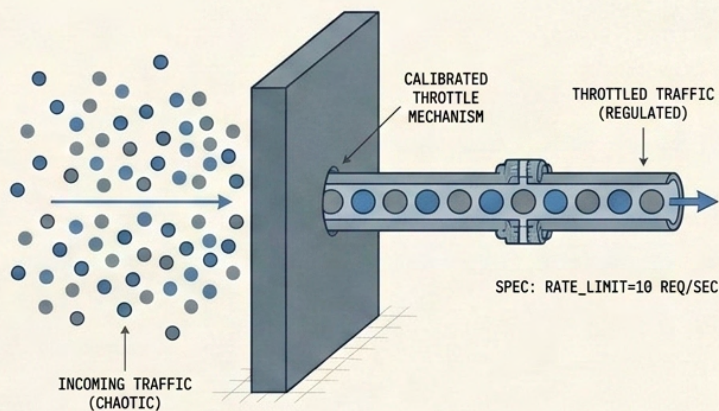
EDGE & TRAFFIC LAYER



CORE FUNCTION:

Throttles incoming requests for a service based on a predefined limit. Acts as a defensive layer preventing excessive use, whether intended (malicious) or unintended (traffic spikes).

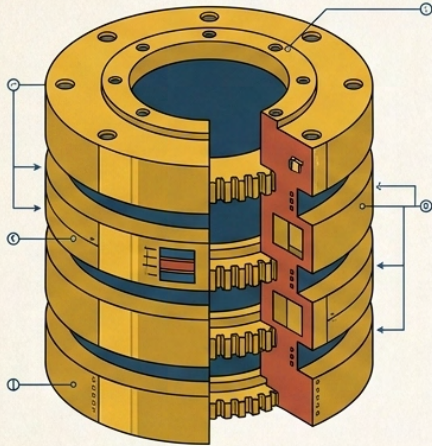
FIRST PRINCIPLE:
The defensive throttle.



SPEC: LIMIT_TYPE=WINDOW-BASED
SPEC: ENFORCEMENT=STRICT
DESIGN PRINCIPLE: TRAFFIC REGULATION & PROTECTION

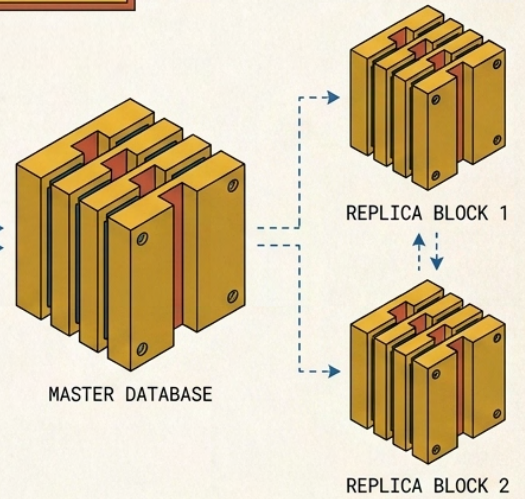
STORAGE & STATE LAYER

STORAGE & STATE LAYER



Stores, retrieves, modifies, and deletes data for complex data-processing procedures. Heavy reliance on replication, partitioning, and distributed analysis for scalability.

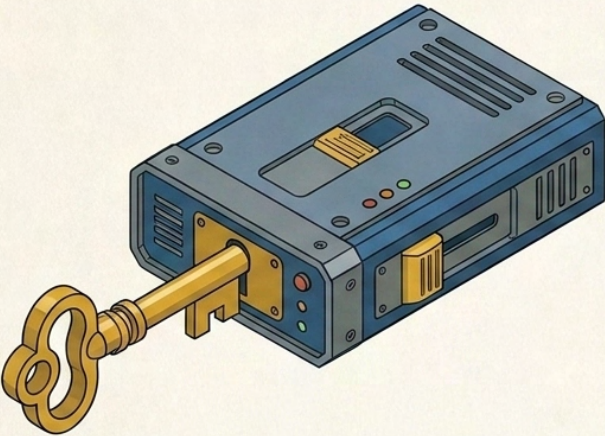
Figure 1 illustrates the data structure, showing three overlapping tables representing different data sources. The tables are labeled 'table', 'table2', and 'table3'. Each table has columns for 'name', 'surname', 'age', 'sex', 'city', and 'country'. The tables are arranged in a 3D space, with 'table' at the bottom, 'table2' in the middle, and 'table3' at the top, illustrating how they are combined into a single dataset.



KEY-VALUE STORE

STORAGE & STATE LAYER

STORAGE & STATE LAYER



CORE FUNCTION:

A non-relational database that stores data purely as **key-value pairs**. Optimized heavily for massive **scalability**, durability, and configurability over complex querying.

FIRST PRINCIPLE:

High-speed, schematic-less lookup.

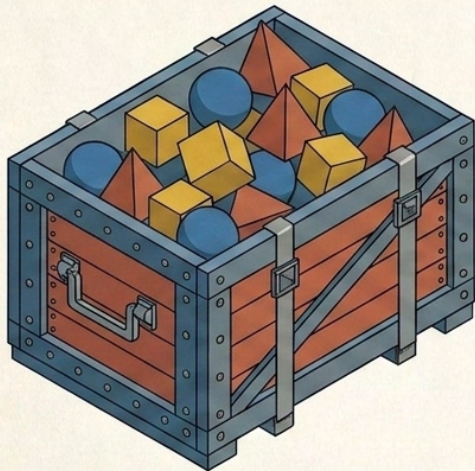
KEY	VALUE
User_123	Session Data
Product_456	{...json_object...}
Sensor_Data_A	Binary_Blob
Config_Flag_1	true
Config_Flag_2	false
Sensor_Data_3	true
Sensor_Data_4	{...json_object...}
Config_Flag_5	Binary_Blob
Config_Flag_6	true
Product_Flag_2	true
Sensor_Data_8	{...json_object...}
Sensor Data 9	true

SPEC: SCHEMA=FLEXIBLE
MAX_KEY_SIZE=1KB
MAX_VALUE_SIZE=10MB

BLOB STORE

STORAGE & STATE LAYER

STORAGE & STATE LAYER



CORE FUNCTION:

A specialized, massively scalable storage solution exclusively designed for unstructured data like multimedia files and binary executables.

FIRST PRINCIPLE:
Mass unstructured storage.



VIDEO

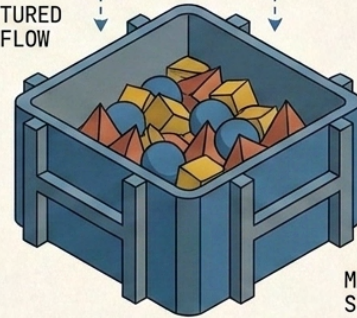


PDFs



BINARIES

UNSTRUCTURED
MEDIA FLOW

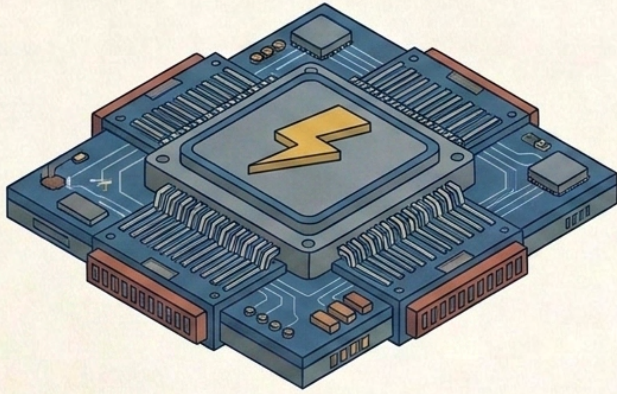


MASSIVE OPEN
STORAGE VAT

DISTRIBUTED CACHING

STORAGE & STATE LAYER

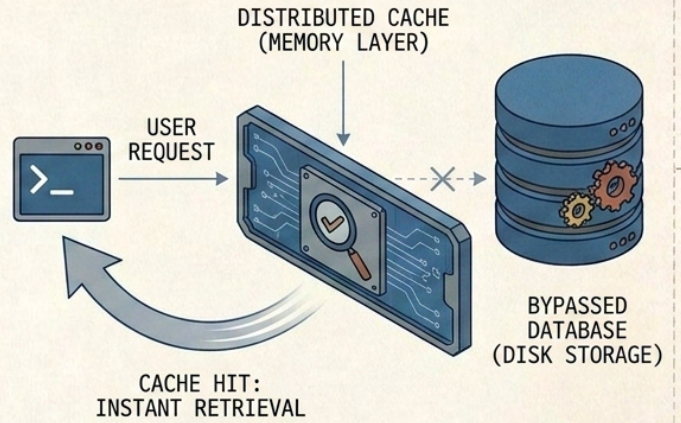
STORAGE & STATE LAYER



CORE FUNCTION:

Multiple coordinating cache servers that store frequently accessed data in memory. Bypasses repetitive, expensive disk reads to deliver near-instantaneous retrieval.

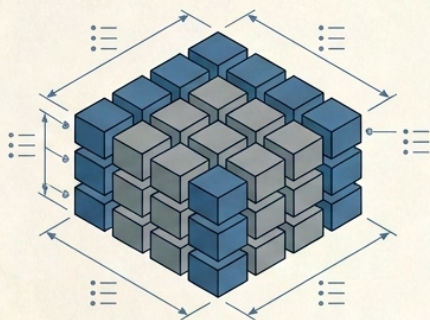
FIRST PRINCIPLE:
The high-speed memory layer.



COMPARISON MATRIX:

The Storage Spectrum

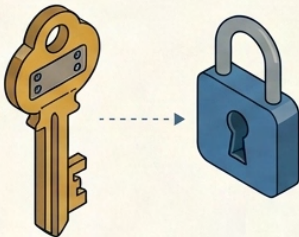
DATABASES (RELATIONAL)



DATA TYPE: Highly structured, relational grids.

PRIMARY USE: Complex queries, ACID transactions, data analysis.

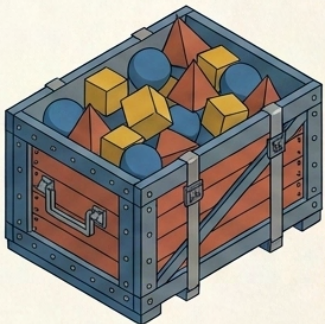
KEY-VALUE STORE



DATA TYPE: Schematic-less paired identifiers.

PRIMARY USE: Ultra-fast lookups, high scalability, session states.

BLOB STORE

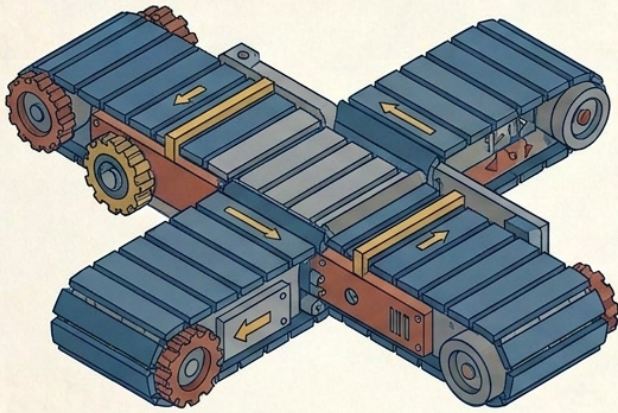


DATA TYPE: Unstructured binary assets.

PRIMARY USE: Multimedia hosting, backups, large executable files.

DISTRIBUTED MESSAGING QUEUE

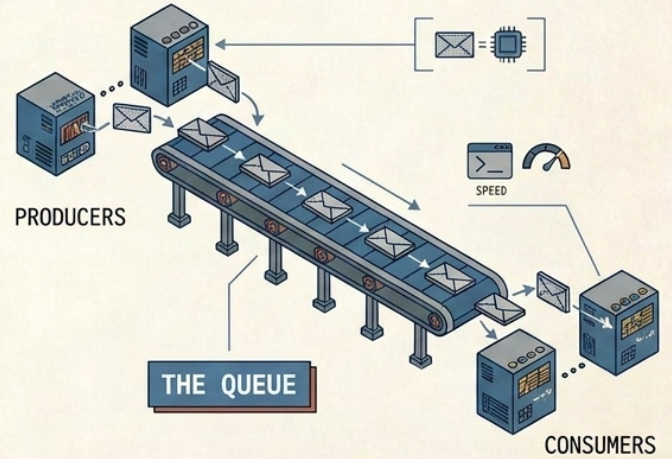
ASYNCHRONOUS LAYER



CORE FUNCTION:

A queue of multiple servers mediating between producers and consumers. Decouples interacting entities, resulting in independent scalability and enhanced system reliability.

FIRST PRINCIPLE: Point-to-point decoupling.



PUBLISH / SUBSCRIBE SYSTEM (PUB-SUB)

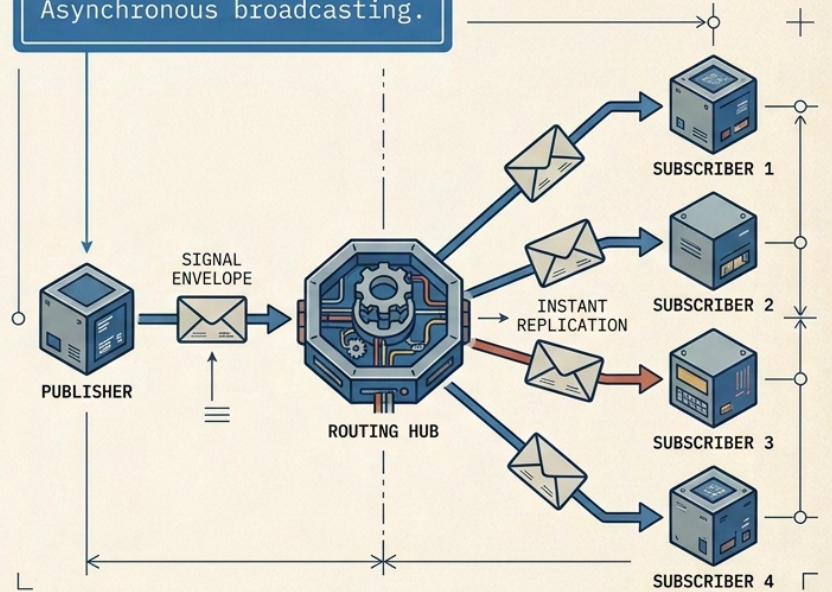
ASYNCHRONOUS LAYER



CORE FUNCTION:

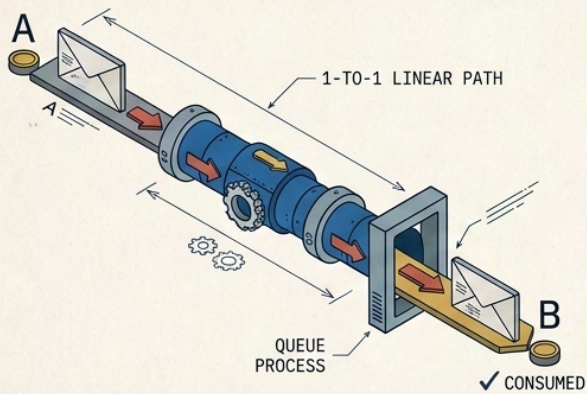
An asynchronous service-to-service communication method. Deeply integrated into serverless architectures, microservices, and large-scale data processing systems.

FIRST PRINCIPLE:
Asynchronous broadcasting.



ROUTING LOGIC: QUEUE vs. PUB/SUB

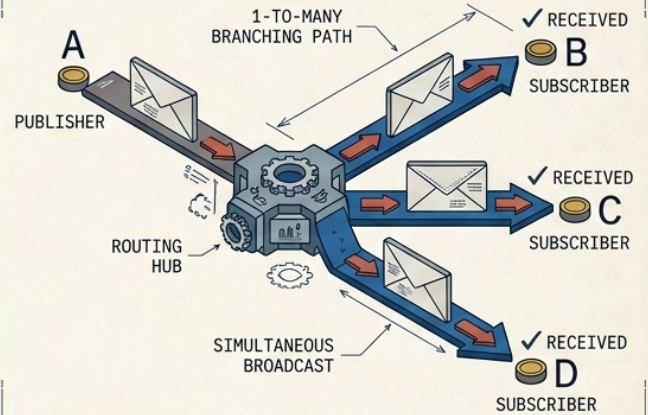
DISTRIBUTED MESSAGING QUEUE



1-to-1 processing.
Work is consumed once and removed from the pipeline.

Primary Goal: Load distribution and pacing.

PUB/SUB SYSTEM

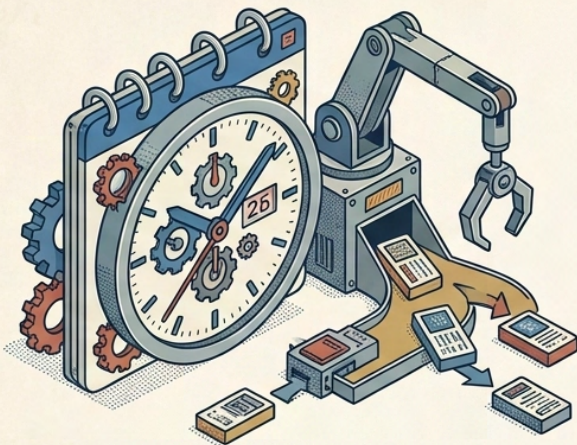


1-to-Many broadcasting.
The exact same event is replicated to all interested subscribers.

Primary Goal: Event notification and parallel reactions.

DISTRIBUTED TASK SCHEDULING

ASYNCHRONOUS LAYER



CORE FUNCTION:

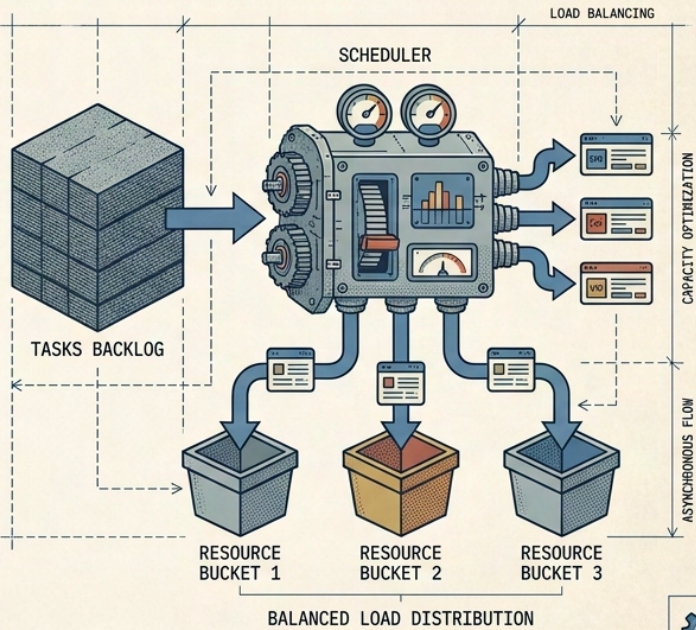
Mediates between tasks and system resources. Intelligently allocates resources to tasks to meet both task-level and system-level goals. Crucial for offloading background processing for asynchronous completion.

TECHNICAL SPECIFICATION



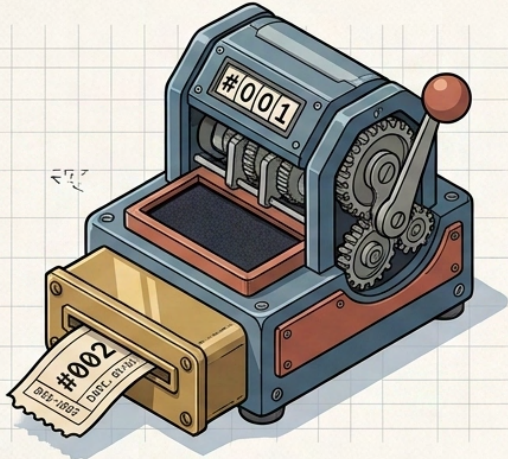
FIRST PRINCIPLE:

Resource mediation & background execution.



SEQUENCER: SPECIALIZED LOGIC LAYER

ASYNCHRONOUS LAYER

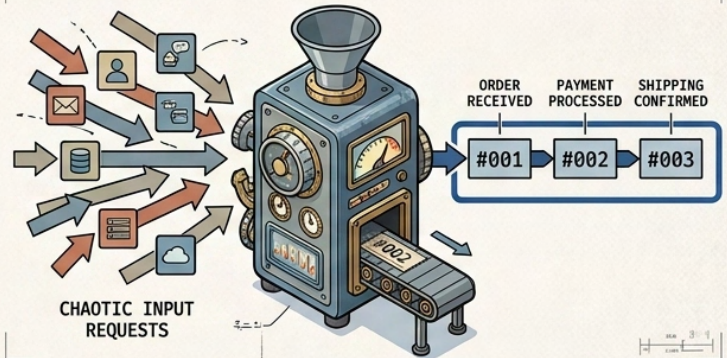


CORE FUNCTION:

A generator for unique IDs across a distributed system, with a major focus on maintaining strict event causality. Uses specific algorithmic methods to ensure no duplicates exist across multiple nodes.

FIRST PRINCIPLE:

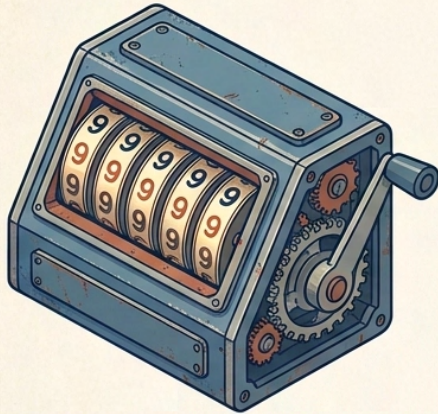
Causal uniqueness.



CAUSAL ORDER & UNIQUE IDENTIFICATION FLOW

SHARDED COUNTERS

SPECIALIZED LOGIC LAYER

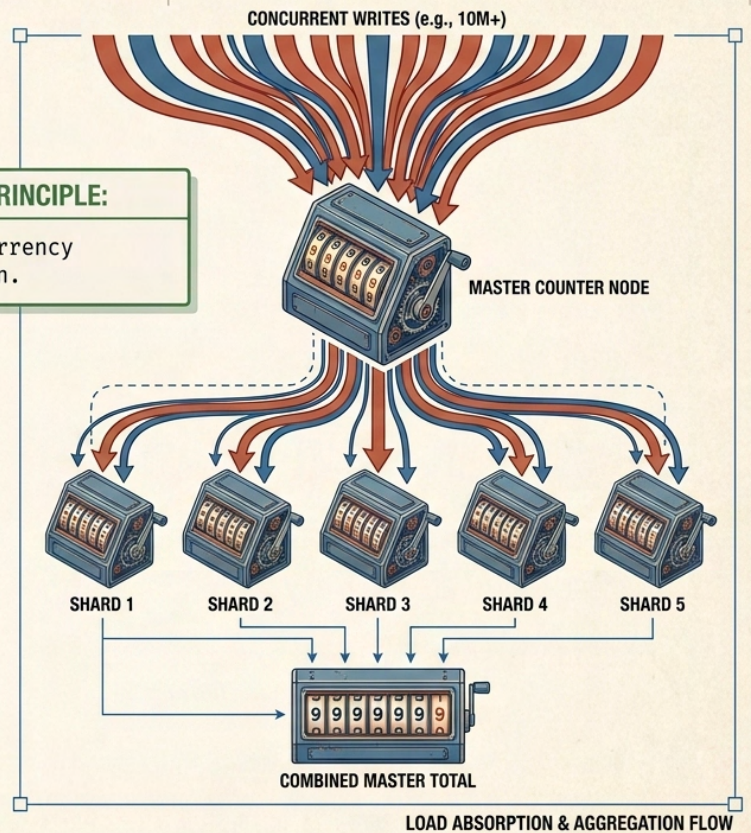


CORE FUNCTION:

An efficient distributed counting system designed to absorb millions of concurrent read/write requests (e.g., likes on a viral event) without locking databases or degrading performance.

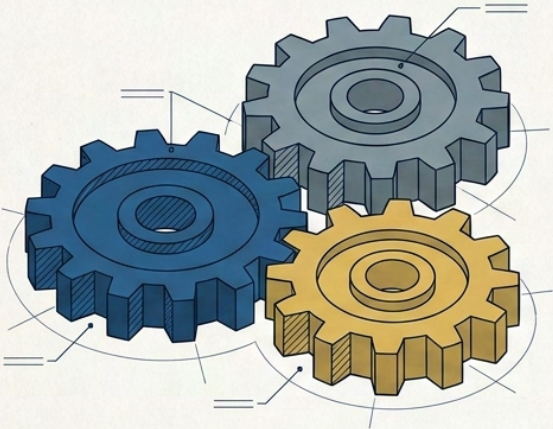
FIRST PRINCIPLE:

High-concurrency aggregation.



DISTRIBUTED SEARCH

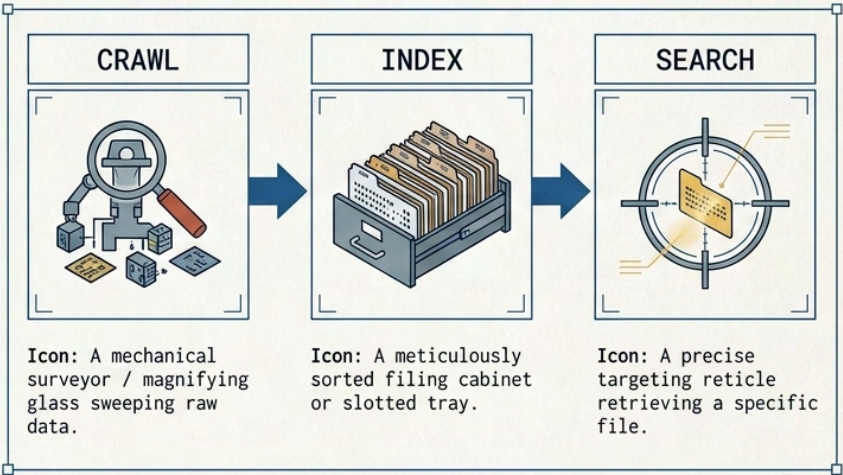
SPECIALIZED LOGIC LAYER



CORE FUNCTION:

Takes a user query and returns relevant content in milliseconds. Built entirely on three integral, sequential components: **crawling data**, **structuring an index**, and **executing the search**.

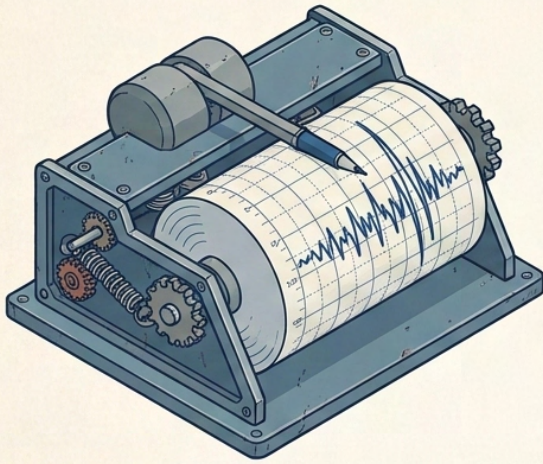
FIRST PRINCIPLE:
Near-instant discovery.



TECHNICAL SPECIFICATION

SERVICE MONITORING

OBSERVABILITY LAYER



CORE FUNCTION:

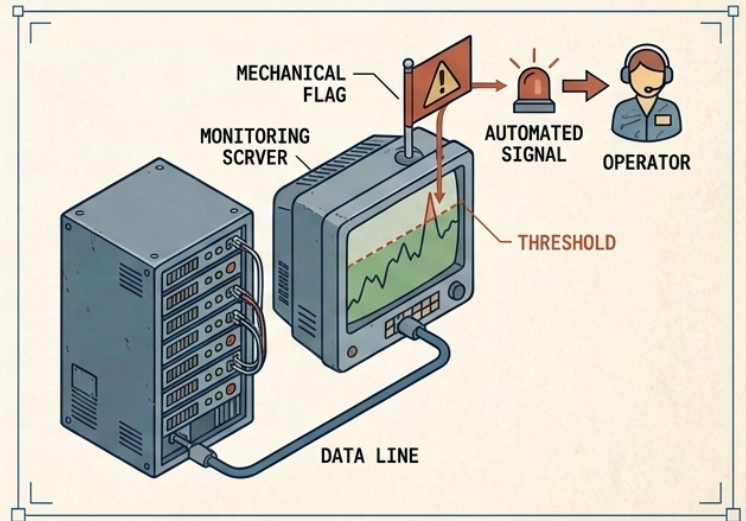
Analyzes system health and tracks both server-side and client-side errors. Alerts stakeholders to anomalies, allowing administrators to act before an impending problem becomes a major issue.



TECHNICAL SPECIFICATION

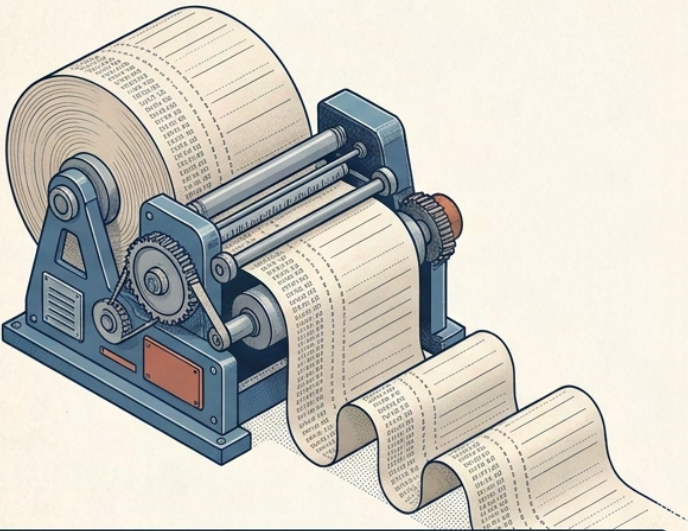


FIRST PRINCIPLE:
The early warning system.



DISTRIBUTED LOGGING

OBSERVABILITY LAYER



CORE FUNCTION:

Logging is inherently an I/O intensive and time-consuming operation. This scalable, reliable system allows distributed services to log their events efficiently without dragging down core application performance.

FIRST PRINCIPLE:
Efficient event historicization.

